Industrial Internship Report on

Agriculture Crop Production Prediction & Smart City Traffic Forecasting

Prepared by

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Executive Summary

This report provides details of the Industrial Internship provided by upskill Campus and The IoT Academy in collaboration with Industrial Partner UniConverge Technologies Pvt Ltd (UCT).

This internship was focused on a project/problem statement provided by UCT. We had to finish the project including the report in 6 weeks' time.

My internship involved completing two real-world machine learning projects: Agriculture Crop Production Prediction and Smart City Traffic Forecasting. These projects focused on applying machine learning techniques to solve real-world prediction problems through data preprocessing, exploratory data analysis, feature engineering, model development, and performance evaluation.

This internship gave me a very good opportunity to get exposure to Industrial problems and design/implement solution for that. It was an overall great experience to have this internship.

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1 Preface

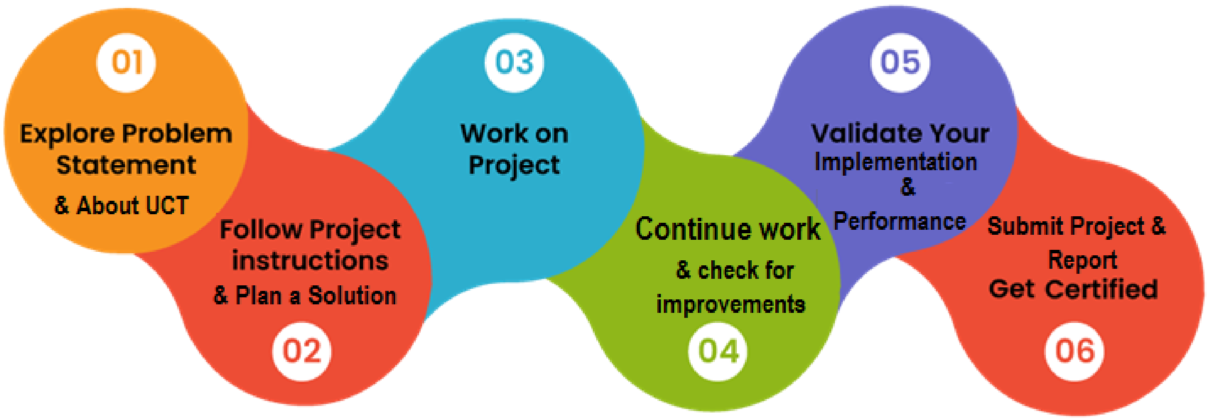
The **Machine Learning and Data Analytics Internship** conducted by **upSkill Campus**, **The IoT Academy**, and **UniConverge Technologies Pvt. Ltd. (UCT)** provided me with an excellent opportunity to gain practical experience in applying machine learning techniques to real-world problems.

During the six-week internship, I completed two machine learning projects: **Agriculture Crop Production Prediction** and **Smart City Traffic Forecasting**. These projects involved understanding the problem statements, exploring and cleaning datasets, performing exploratory data analysis (EDA), statistical analysis, feature engineering, developing predictive models using **Linear Regression** and **Random Forest Regression**, and evaluating model performance using standard metrics.

The internship was well-structured, beginning with project selection and dataset exploration, followed by data preprocessing, statistical analysis, machine learning model development, and final documentation. This step-by-step approach helped me understand the complete machine learning workflow from raw data to model evaluation.

Throughout the internship, I improved my technical knowledge of Python, Pandas, NumPy, Matplotlib, SciPy, and Scikit-learn. In addition to technical skills, I enhanced my analytical thinking, problem-solving abilities, and documentation skills, which will be valuable in my academic and professional career.

I sincerely thank **upSkill Campus**, **The IoT Academy**, **UniConverge Technologies Pvt. Ltd. (UCT)**, my college, and everyone who supported and guided me throughout this internship. I hope this report serves as a useful summary of the knowledge and practical experience I gained during the program.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2 Introduction

2.1 About UniConverge Technologies Pvt Ltd

A company established in 2013 and working in Digital Transformation domain and providing Industrial solutions with prime focus on sustainability and RoI.

For developing its products and solutions it is leveraging various **Cutting Edge Technologies e.g. Internet of Things (IoT), Cyber Security, Cloud computing (AWS, Azure), Machine Learning, Communication Technologies (4G/5G/LoRaWAN), Java Full Stack, Python, Front end** etc.

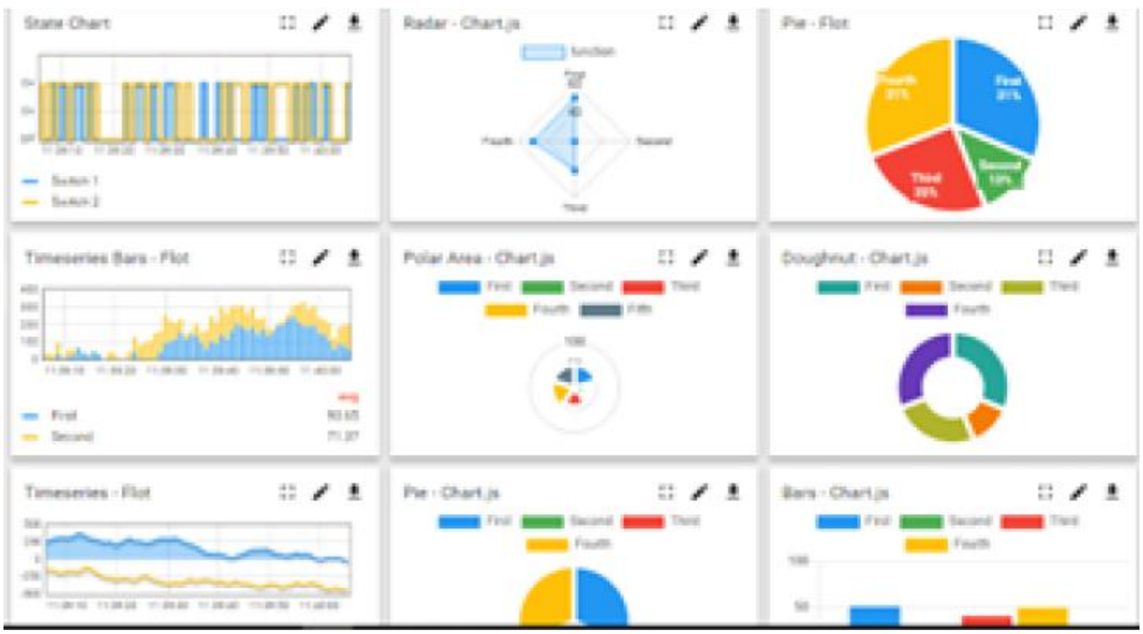
i. UCT IoT Platform (uct Insight)

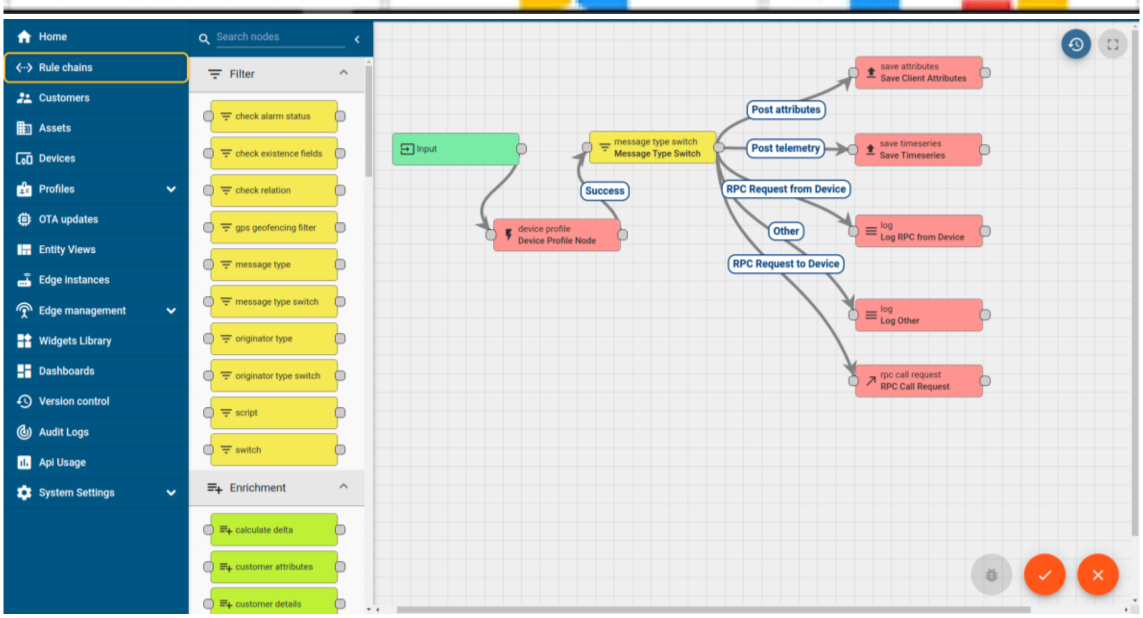
UCT Insight is an IOT platform designed for quick deployment of IOT applications on the same time providing valuable “insight” for your process/business. It has been built in Java for backend and ReactJS for Front end. It has support for MySQL and various NoSql Databases.

- It enables device connectivity via industry standard IoT protocols - MQTT, CoAP, HTTP, Modbus TCP, OPC UA
- It supports both cloud and on-premises deployments.

It has features to

• Build Your own dashboard
• Analytics and Reporting
• Alert and Notification
• Integration with third party application(Power BI, SAP, ERP)
• Rule Engine





ii. Smart Factory Platform (FACTORY WATCH)

Factory watch is a platform for smart factory needs.

It provides Users/ Factory

- with a scalable solution for their Production and asset monitoring
- OEE and predictive maintenance solution scaling up to digital twin for your assets.
- to unleased the true potential of the data that their machines are generating and helps to identify the KPIs and also improve them.
- A modular architecture that allows users to choose the service that they what to start and then can scale to more complex solutions as per their demands.

Its unique SaaS model helps users to save time, cost and money.

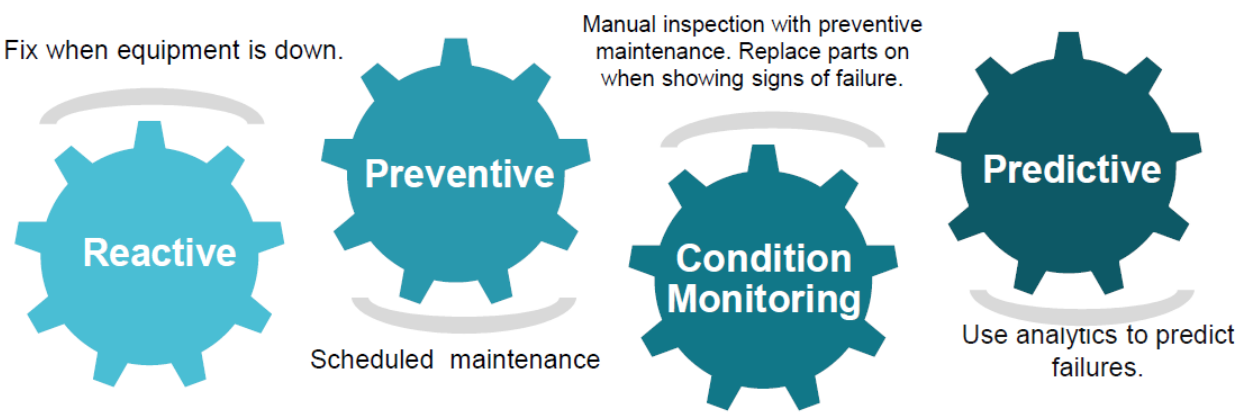
Machine	Operator	Work Order ID	Job ID	Job Performance	Job Progress		Output		Rejection	Time (mins)				Job Status	End Customer
					Start Time	End Time	Planned	Actual		Setup	Pred	Downtime	Idle		
CNC_S7_81	Operator 1	WO0405200001	4168	58%	10:30 AM		55	41	0	80	215	0	45	In Progress	i
CNC_S7_81	Operator 1	WO0405200001	4168	58%	10:30 AM		55	41	0	80	215	0	45	In Progress	i

iii. LoRaWAN based Solution

UCT is one of the early adopters of LoRAWAN teschnology and providing solution in Agritech, Smart cities, Industrial Monitoring, Smart Street Light, Smart Water/ Gas/ Electricity metering solutions etc.

iv. Predictive Maintenance

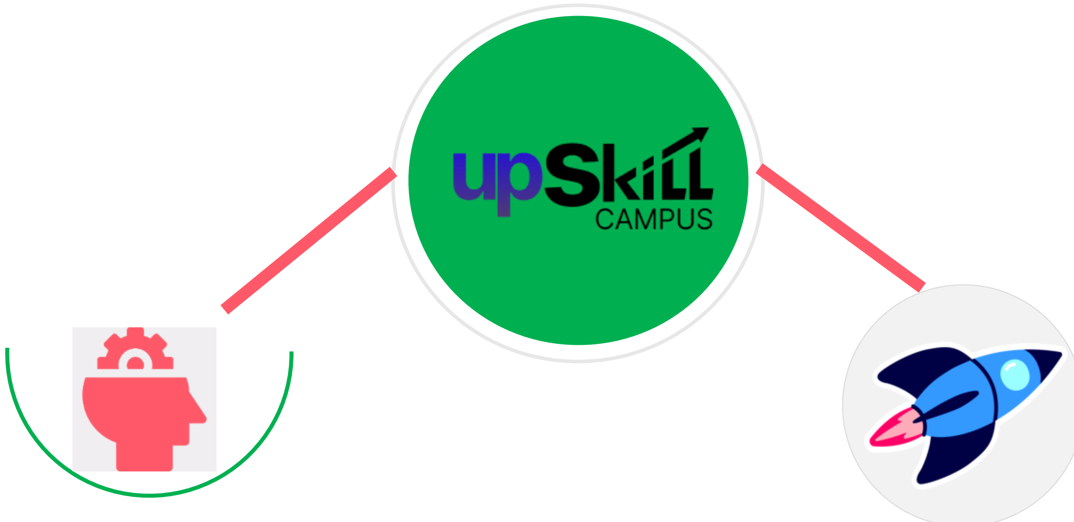
UCT is providing Industrial Machine health monitoring and Predictive maintenance solution leveraging Embedded system, Industrial IoT and Machine Learning Technologies by finding Remaining useful life time of various Machines used in production process.



2.2 About upskill Campus (USC)

upskill Campus along with The IoT Academy and in association with Uniconverge technologies has facilitated the smooth execution of the complete internship process.

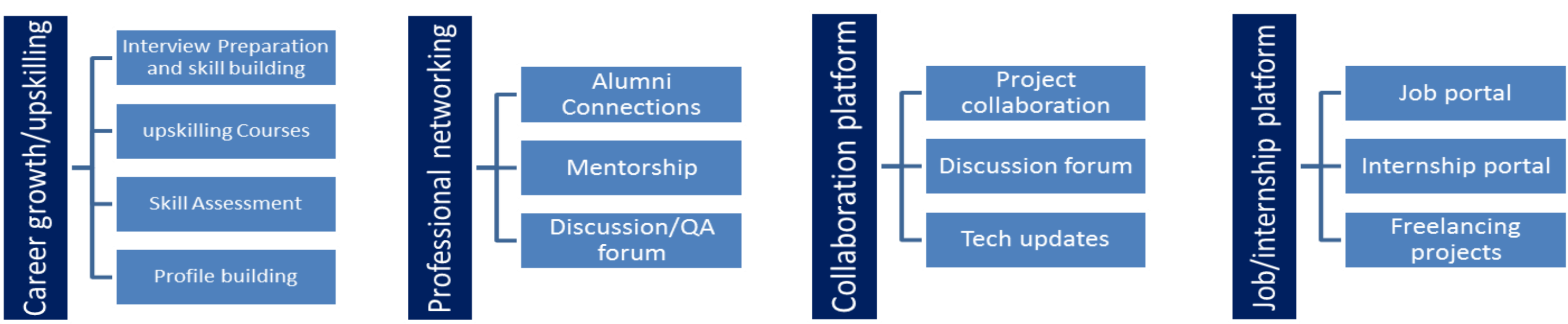
USC is a career development platform that delivers **personalized executive coaching** in a more affordable, scalable and measurable way.



Seeing need of upskilling in self paced manner along-with additional support services e.g. Internship, projects, interaction with Industry experts, Career growth Services

upSkill Campus aiming to upskill 1 million learners in next 5 year

https://www.upskillcampus.com/



2.3 The IoT Academy

The IoT academy is EdTech Division of UCT that is running long executive certification programs in collaboration with EICT Academy, IITK, IITR and IITG in multiple domains.

2.4 Objectives of this Internship program

The objective for this internship program was to

☛ get practical experience of working in the industry.

☛ to solve real world problems.

☛ to have improved job prospects.

☛ to have Improved understanding of our field and its applications.

☛ to have Personal growth like better communication and problem solving.

2.5 Reference

[1] https://scikit-learn.org/

[2] https://matplotlib.org/

[3] https://pandas.pydata.org/

2.6 Glossary

Terms	Acronym
EDA	**Exploratory Data Analysis**
MAE	**Mean Absolute Error**
RMSE	**Root Mean Squared Error**
ML	**Machine Learning**
R^2	**Coefficient of Determination**

3 Problem Statement

3.1 Agriculture Crop Production Prediction

Agriculture plays a vital role in India's economy, and accurate prediction of crop production can help farmers and policymakers make informed decisions. However, crop yield depends on multiple factors such as cultivation cost, production cost, crop type, and geographical location. Analyzing these factors manually is difficult and often leads to inefficient planning. The objective of this project was to develop a machine learning model capable of predicting crop production based on historical agricultural data, thereby supporting better agricultural planning and resource management.

3.2 Smart City Traffic Forecasting

Rapid urbanization has resulted in increasing traffic congestion in smart cities, making traffic management a major challenge. Predicting vehicle traffic in advance helps improve traffic control, route planning, and infrastructure utilization. The objective of this project was to analyze historical traffic data and develop a machine learning model capable of forecasting vehicle traffic using temporal and location-based features, enabling more efficient traffic management.

4 Existing and Proposed solution

Existing Solutions

For **Agriculture Crop Production Prediction**, traditional prediction methods often rely on manual analysis, statistical reports, or expert judgment. These methods may not effectively capture the complex relationships between multiple factors influencing crop production and can be time-consuming.

For **Smart City Traffic Forecasting**, conventional traffic management systems mainly depend on historical averages or fixed traffic control strategies. Such approaches may not accurately adapt to changing traffic patterns or provide reliable future traffic predictions.

Limitations of Existing Solutions

- Manual analysis is time-consuming and prone to human error.
- Traditional statistical methods have limited predictive capability for complex datasets.
- Existing approaches often fail to capture non-linear relationships among variables.
- Conventional traffic management systems may not respond effectively to changing traffic conditions.

Proposed Solution

To address these limitations, two machine learning models were developed.

The **Agriculture Crop Production Prediction** project predicts crop yield using agricultural features such as crop type, state, cultivation cost, and production cost. The workflow included data preprocessing, exploratory data analysis, statistical analysis, feature engineering, and regression model development.

The **Smart City Traffic Forecasting** project predicts future traffic volume using historical traffic data and temporal features such as hour, day of the week, month, year, and junction information. Similar preprocessing and machine learning techniques were applied.

For both projects, **Linear Regression** was implemented as the baseline model, while **Random Forest Regression** was used to improve prediction accuracy. Model performance was evaluated using **R^2 Score, Mean Absolute Error (MAE), and Root Mean Squared Error (RMSE).**

Value Addition

The proposed solution offers the following advantages:

- Improved prediction accuracy using machine learning algorithms.
- Better support for data-driven decision-making.
- Efficient analysis of large datasets through automated prediction models.
- A scalable workflow that can be extended with larger datasets or advanced machine learning techniques in future.

4.1 Code submission (Github link)

https://github.com/Abi0507/upskillcampus

4.2 Report submission (Github link) :

https://github.com/Abi0507/upskillcampus/blob/main/MachineLearningProjects_Abinaya_S_USC_UCT.pdf

5 Proposed Design/ Model

The proposed solution follows a structured machine learning workflow for both projects. The process begins with collecting and understanding the dataset, followed by data preprocessing to improve data quality. Exploratory Data Analysis (EDA) and statistical analysis are performed to identify patterns, relationships, and important features. Feature engineering is then applied to prepare the data for machine learning models. Two regression models, Linear Regression and Random Forest Regression, are trained and evaluated using standard performance metrics. Finally, feature importance analysis is performed to identify the variables that have the greatest influence on prediction.

5.1 High Level Diagram (if applicable)

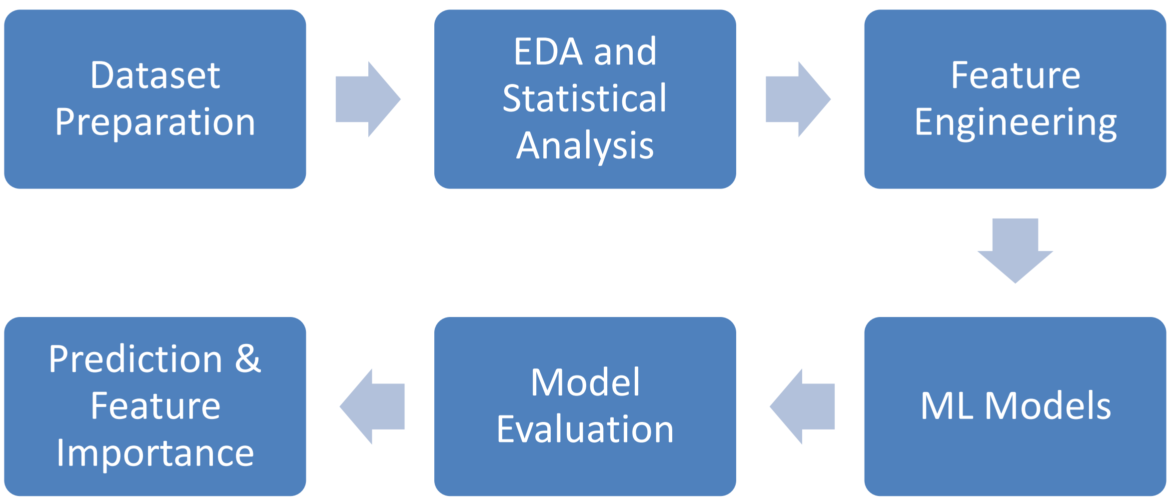
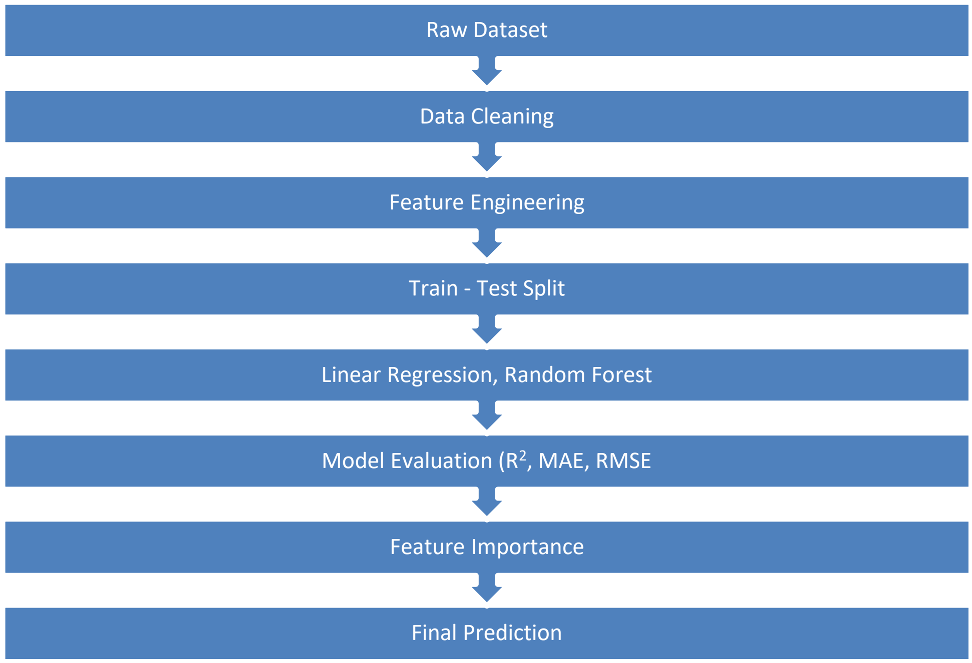


Figure 1: HIGH LEVEL DIAGRAM OF THE SYSTEM

5.2 Low Level Diagram (if applicable)



5.3 Interfaces (if applicable)

The projects were implemented using **Jupyter Notebook** as the development environment. The interface consisted of Python scripts for data preprocessing, visualization, machine learning model development, and performance evaluation. Various plots such as histograms, scatter plots, correlation heatmaps, and feature importance graphs were used to analyze the datasets and interpret model performance.

6 Performance Test

Before evaluating the machine learning models, the datasets were divided into training and testing sets. Linear Regression was implemented as the baseline model, and Random Forest Regression was developed to improve prediction accuracy. The models were evaluated using **R^2 Score**, **Mean Absolute Error (MAE)**, and **Root Mean Squared Error (RMSE)** to measure their predictive performance.

6.1 Test Plan/ Test Cases

Test Case	Objective	Expected Result
Data Preprocessing	Verify that the datasets are cleaned and suitable for model training	Clean and consistent datasets
Linear Regression	Build a baseline prediction model	Generate baseline performance metrics
Random Forest Regression	Improve prediction accuracy	Higher R^2 and lower MAE & RMSE
Model Evaluation	Compare model performance	Identify the better-performing model

6.2 Test Procedure

- Loaded and preprocessed the datasets.
- Performed exploratory data analysis and statistical analysis.
- Applied feature engineering techniques.
- Split the datasets into training and testing sets.
- Trained Linear Regression and Random Forest Regression models.
- Generated predictions using both models.
- Evaluated the models using R^2 Score, MAE, and RMSE.
- Compared the performance of both models and analyzed feature importance.

6.3 Performance Outcome

Agriculture Crop Production Prediction

Model	R^2 Score	MAE	RMSE
Linear Regression	0.9066	66.93	91.86
Random Forest Regression	0.9567	23.74	62.55

Observation:

The Random Forest Regression model outperformed Linear Regression by achieving a higher R^2 Score and significantly reducing the prediction errors (MAE and RMSE), making it the most suitable model for crop production prediction

Smart City Traffic Forecasting

Model	R^2 Score	MAE	RMSE
Linear Regression	0.6285	11.49	16.59
Random Forest Regression	0.9160	4.80	7.89

Observation:

Random Forest Regression achieved substantially better performance than Linear Regression, demonstrating its ability to capture complex traffic patterns and produce more accurate traffic predictions.

7 My learnings

The internship provided me with valuable practical experience in developing end-to-end machine learning solutions. I learned how to approach real-world problems by understanding datasets, performing data cleaning, exploratory data analysis (EDA), statistical analysis, and feature engineering before building predictive models.

Through the **Agriculture Crop Production Prediction** and **Smart City Traffic Forecasting** projects, I gained hands-on experience in implementing **Linear Regression** and **Random Forest Regression** using Python and Scikit-learn. I also learned how to evaluate model performance using metrics such as **R^2 Score, Mean Absolute Error (MAE), and Root Mean Squared Error (RMSE)** and compare different machine learning models to identify the most suitable solution.

In addition to improving my technical skills in Python, Pandas, NumPy, Matplotlib, SciPy, and Scikit-learn, this internship strengthened my analytical thinking, problem-solving ability, documentation skills, and understanding of the complete machine learning development lifecycle. These learnings will be valuable for future academic projects, internships, and my professional career.

8 Future work scope

The developed machine learning models can be further enhanced by using larger and more diverse datasets to improve prediction accuracy and model generalization.

For the **Agriculture Crop Production Prediction** project, additional factors such as weather conditions, rainfall, soil characteristics, and fertilizer usage can be incorporated to improve crop yield prediction.

For the **Smart City Traffic Forecasting** project, integrating real-time traffic sensor data, weather information, and public event schedules can further improve forecasting performance.

Future work may also include implementing advanced machine learning and deep learning techniques, performing hyperparameter optimization, deploying the models as web applications, and integrating interactive dashboards for real-time visualization and decision support.